

Abindas P

Pune, Maharashtra, India

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SUMMARY

Sophomore Computer Engineering student at AIT Pune (8.82 CGPA) specializing in Full Stack Development and AI-driven System Design. Lead Developer for national-level award-winning platforms (SIH '25 Finalist). Expert in architecting scalable backends, implementing RAG-based LLM services, building optimization engines, and leveraging AWS serverless cloud infrastructure to solve real-world challenges.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: FastAPI, Next.js, Node.js, React.js, PyTorch, Tailwind CSS

Tools/Cloud: AWS (Bedrock, Lambda, Textract, CDK, Amplify), Docker, Redis, MongoDB, PostgreSQL, OR-Tools

Concepts: System Design, Serverless Architecture, RAG (LLMs), Optimization Algorithms, RBAC, REST APIs

PROJECTS

KMRL Operations Decision Support Platform (YatraMind)

SIH '25 Finalist Project

Lead Full Stack Developer

Pune, India

Architected an AI/ML-driven planning platform for **Kochi Metro Rail Ltd** using **FastAPI** and **MongoDB** to automate complex train induction workflows and fleet management.

Engineered a constraint-based optimization engine using **Google OR-Tools** and a custom **Durable Rules Engine**, reducing manual fleet allocation time by **approx. 40%**.

Developed **PyTorch** ML Risk Prediction models to forecast maintenance needs and integrated a secure **RBAC** system with IP tracking for supervisor overrides and operational security.

Implemented time-series telemetry visualization with **InfluxDB** and real-time updates via **Socket.io**, maintaining a complete audit trail of all manual adjustments and approvals.

Saarthi.AI: AI Civic Assistant ([Click to View App](#))

AI for Bharat Hackathon (AWS)

Full Stack Cloud Developer

Pune, India

Architected a production-ready, multilingual civic assistant using **Next.js** and **AWS CDK**, deployed seamlessly and globally via **AWS Amplify**.

Engineered a highly-available **RAG** pipeline with **AWS Bedrock** for querying complex government schemes in English, Hindi, and Marathi.

Integrated **AWS Textract** to process unstructured policy PDFs, and **Amazon Transcribe & Polly** to enable real-time, interactive voice-based accessibility.

Designed event-driven backend workflows utilizing **Python AWS Lambdas** to automate community grievance handlers securely.

DecodX-LLM: Intelligent Query-Retrieval System

HackRx 2025 (Bajaj Finserv Health)

Backend Developer & Innovator

Pune, India

Built a modular **RAG** service utilizing **Gemini 1.5-Flash** and **FAISS** for high-performance context-aware analysis of unstructured document datasets and policy queries.

Designed a hybrid retrieval pipeline using **BM25** and **Vector Embeddings** to handle complex natural language queries with high semantic accuracy and flight number agent integration.

Optimized response generation with custom **FastAPI** middleware for API token validation and efficient document chunking/text extraction under strict hackathon time constraints.

Ranked in the **Top 100 teams** (Final Table) globally, validating the technical feasibility and business impact of the solution for Bajaj Finserv Health.

EDUCATION

Army Institute of Technology (AIT)

Pune, India

Bachelor of Engineering in Computer Engineering (Current CGPA: 8.82)

July 2024 – 2028

Core Subjects: Data Structures & Algorithms (C++), Object Oriented Programming, DBMS, OS.

Kendriya Vidyalaya

India

Higher Secondary Education (Class XII) — PCM Score: 98.34%

Completed 2024

Aggregate: 97.8%. Recipient of Academic Excellence Award for top percentile rankings.

ACHIEVEMENTS

Smart India Hackathon (SIH) 2025 Finalist: Technical excellence in Ministry of Housing challenges.

HackRx 2025 Finalist: Secured a final table spot among thousands of participants nationwide.

LEADERSHIP & EXPERIENCE

Member, Magazine Board

Pune, India

Army Institute of Technology (AIT)

2024 – 2025

Collaborated with the editorial board to curate content and managed production timelines for the annual publication.

Technical Community Member

Pune, India

Independent Developer & Open Source Enthusiast

July 2024 – Present

Upskilling in modular C++ and sharing knowledge on System Design and AI via open-source platforms.